

Problem Set 2Question 1

(a)

$$Z = C + I + G + X - M$$

$$Z = C_0 + c_1(Y - T) + d_0 + d_1Y + G + x_1Y^* - m_1Y$$

At eq^m, then

$$Y = Z$$

$$= C_0 + c_1(Y - T) + d_0 + d_1Y + G + x_1Y^* - m_1Y$$

$$= C_0 + d_0 + G - c_1T + x_1Y^* + (c_1 + d_1 - m_1)Y$$

$$(1 - c_1 - d_1 + m_1)Y = C_0 + d_0 + G - c_1T + x_1Y^*$$

$$\text{At eq}^m, Y = \frac{1}{1 - c_1 - d_1 + m_1} (C_0 + d_0 + G) - \frac{c_1}{1 - c_1 - d_1 + m_1} T + \frac{1}{1 - c_1 - d_1 + m_1} x_1Y^*$$

Note that $\frac{1}{1 - c_1 - d_1 + m_1}$ is the multiplier and $-\frac{c_1}{1 - c_1 - d_1 + m_1}$ is the tax

multiplier.

(b) Suppose government purchase ↑ by 1 unit, then $\Delta G = 1$

$$\Delta Y = \frac{1}{1 - c_1 - d_1 + m_1} \Delta G$$

Note that the slope of Z in closed economy is $c_1 + d_1$

and the slope of Z in open economy is $c_1 + d_1 - m_1$ where d_1 captures the effect of additional unit of income on investment and m_1 captures the effect of additional unit of income on imports. $d_1 \uparrow \rightarrow$ multiplier $\uparrow$ but $m_1 \uparrow \rightarrow$ multiplier $\downarrow$.

(c) As net export, $Nx = x_1Y^* - m_1Y$, then

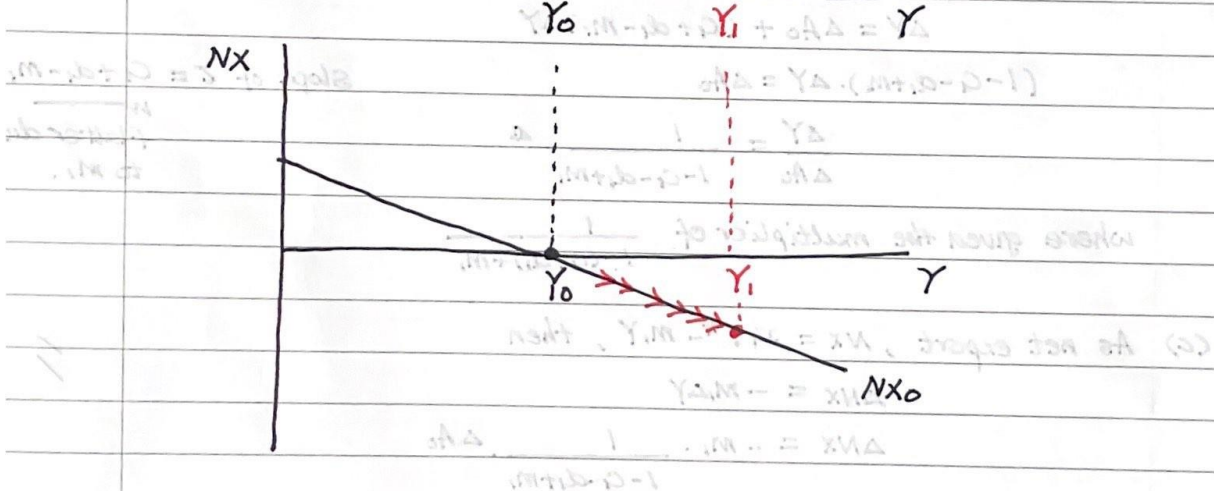
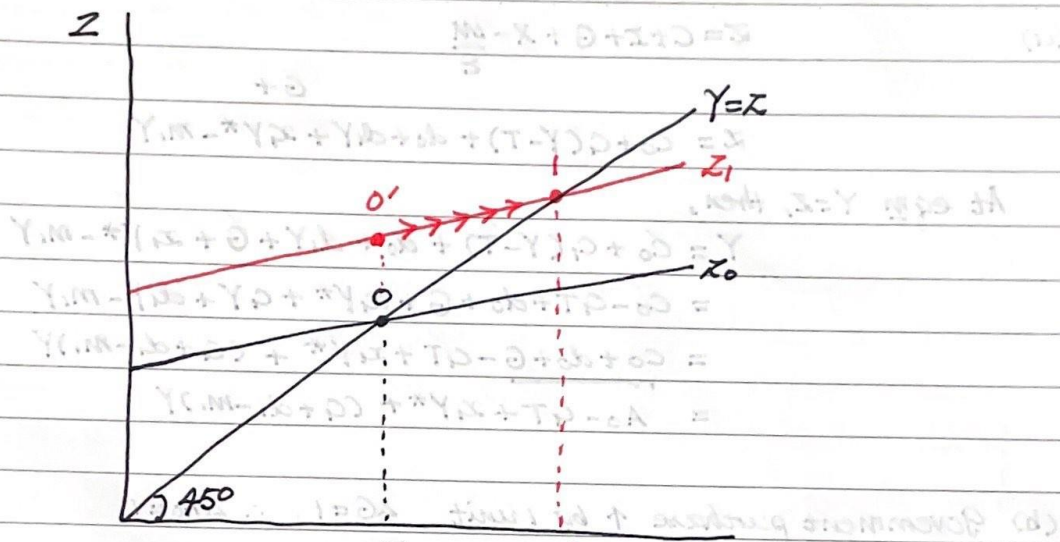
$$\Delta Nx = x_1 \Delta Y^* - m_1 \Delta Y$$

Assuming that $\Delta Y^* = 0$, then ΔNx can be written as

$$\Delta Nx = -m_1 \Delta Y$$

$$\Delta Nx = -m_1 \left(\frac{1}{1 - c_1 - d_1 + m_1} \right) \Delta G$$

When $G \uparrow \rightarrow Z \uparrow \rightarrow$ Since $Y = Z$, $Y \uparrow \rightarrow M \uparrow$. With $\bar{X}$, $\bar{M} \uparrow$, $Nx = \bar{X} - \bar{M} \rightarrow Nx \downarrow$



Note that there is a typo in the assumption of 1(b) from the question set. The assumption should be $0 < m_1 < c_1 + d_1 < 1$. It ensures that $0 < (1 - c_1 - d_1 + m_1) < 1$ so the multiplier is defined, positive, and greater than 1

Question 2

(a) 2 economies : $m_1^S = 0.5$ and $m_1^L = 0.1$. Both have $C_1 + d_1 = 0.6$

Between the two economies, one is larger. We would expect the larger economy to produce more variety of goods as compared to the small economy. Hence m_1 of a larger economy is likely to be smaller $\therefore m_1^L < m_1^S$ where the superscript of L is for larger economy and S is for small economy.

(b) The multiplier is $\frac{1}{1 - (C_1 + d_1) + m_1} = k$ $\Delta Y = k \cdot \Delta G$

and the change in NX is $\Delta NX = -m_1 \cdot \Delta Y$.

	Small economy	Large economy
Multiplier, k	$k^S = \frac{1}{1 - (C_1 + d_1) + m_1^S}$	$k^L = \frac{1}{1 - (C_1 + d_1) + m_1^L}$
	$k^S = \frac{1}{1 - 0.6 + 0.5}$	$k^L = \frac{1}{1 - 0.6 + 0.1}$
	$= \frac{1}{0.9}$	$= \frac{1}{0.5}$
	$k^S = 1.1$	$k^L = 2$

Change in NX, $\Delta NX^S = -m_1^S \cdot \Delta Y$ $\Delta NX^L = -m_1^L \cdot \Delta Y$

$\Delta NX = -0.5 \Delta Y$ $= -0.1 \Delta Y$

If $\Delta G = 1$, then, $\Delta NX^S = -0.5 \cdot k^S \cdot \Delta G$ $\Delta NX^L = -0.1 \cdot k^L \cdot \Delta G$

$$= -0.5(1.1)(1) = -0.1(2)(1)$$

$$= -0.55 \quad \Delta NX = -0.2$$

(c) Larger effect on output : Larger economy

Larger effect on NX : Small economy

Question 3

$$C = 10 + 0.8(Y - T)$$

$$I = 10$$

$$G = 10$$

$$T = 10$$

$$M = 0.3Y$$

$$X = 0.3Y^*$$

(a) Given Y^* , $Z = C + I + G + X - M$

$$= 10 + 0.8(Y - 10) + 10 + 10 + 0.3Y^* - 0.3Y$$

$$Z = 22 + 0.5Y + 0.3Y^*$$

At eq^m $Y = Z$

$$Y = 22 + 0.5Y + 0.3Y^*$$

$$(1 - 0.5) Y = 22 + 0.3Y^*$$

$$0.5Y = 22 + 0.3Y^*$$

$$Y = 2(22) + 2(0.3)Y^*$$

$\nwarrow$
multiplier

$$Y = 44 + 0.6Y^*$$

In this open economy, the multiplier is $\frac{1}{1 - c + m_1} = \frac{1}{1 - 0.8 + 0.3} = \frac{1}{0.5} = 2$

If the economy is closed, the multiplier is $\frac{1}{1 - c} = \frac{1}{1 - 0.8} = \frac{1}{0.2} = 5$.

In the open economy, some of the increase in autonomous expenditure (say G) falls on foreign goods, so the multiplier is smaller.

(b) Note that the domestic economy is characterized by the following eq^m

$$Y = 44 + 0.6Y^*$$

And by symmetry, the foreign economy has the similar eq^m condition

$$Y^* = 44 + 0.6Y$$

The equilibrium output in domestic economy is

$$Y = 44 + 0.6(44 + 0.6Y)$$

$$\therefore Y = 110$$

Since the two economies are identical, then

$$Y^* = 110$$

Note that $Z = 22 + 0.5Y + 0.3Y^*$ and $Y^* = 44 + 0.6Y$

then, $Z = 22 + 0.5Y + 0.3(44 + 0.6Y)$

$$Z = 35.2 + 0.68Y$$

At eq^m $Y = Z$

$$Y = 35.2 + 0.68Y$$

$$(1 - 0.68)Y = 35.2$$

$$Y = \frac{1}{1 - 0.68} \cdot 35.2$$

multiplier

$$Y = 3.125(35.2) \quad \therefore k = 3.125$$

The multiplier is higher than 2 (open economy) b/c it takes into account the fact that an increase in domestic income leads to higher foreign income. The increase in foreign income then leads to an increase in domestic exports.

(c) Assume that target output $Y_f = 125$ and G^* is fixed.

Since $Y = 110$, and $Y_f = 125$, then to achieve $Y_f = 125$,

$$\Delta Y = 125 - 110 = 15$$

This means that with multiplier $K = 3.125$, to achieve $Y \uparrow$ of 15

$$\Delta Y = K \cdot \Delta G$$

$$15 = 3.125(\Delta G) \quad \therefore \Delta G = \frac{15}{3.125} = 4.8$$

We need G to increase by 4.8.

Domestic economy, $NX = 0.3Y^* - 0.3Y_f$. We know $Y_f = 125$ but not new Y^* after $G \uparrow$ of 4.8.

To find Y^* when $Y = Y_f = 125$, we sub this into Y^*

$$\begin{aligned} Y^* &= 44 + 0.6Y \\ &= 44 + 0.6(125) \end{aligned}$$

$$Y^* = 119$$

Note that the fiscal policy conducted in domestic economy will spillover to foreign economy with Y^* increases from 110 to 119.

With $Y_f = 125$ and $Y^* = 119$

$$NX = 0.3Y^* - 0.3Y_f$$

$$= 0.3(119) - 0.3(125)$$

$$NX = -1.8$$

$$NX^* = 0.3Y - 0.3Y^*$$

$$= 0.3(125) - 0.3(119)$$

$$NX^* = 1.8$$

Budget in domestic economy,

$$T - G = 10 - (G + \Delta G)$$

$$= 10 - (10 + 4.8)$$

$$= -4.8$$

Fiscal deficit of 4.8

Budget in foreign economy

$$T^* - G^* = 10 - 10 = 0$$

Balanced budget.

(d) Suppose $Y_f = Y_f^* = 125$ and $\Delta G = \Delta G^*$

Then $Y_f = 10 + 0.8(Y_f - 10) + 10 + G_1 + 0.3Y_f^* - 0.3Y_f$

$$125 = 10 + 0.8(125 - 10) + 10 + G_1 + 0.3(125) - 0.3(125)$$

$$G_1 = 125 - 112 = 13$$

This means that $\Delta G = \Delta G^* = 3$ Note: This increase in G is smaller than if domestic economy were to raise

And net export, NX

$$NX = 0.3Y_f^* - 0.3Y_f$$

$$= 0.3(125) - 0.3(125)$$

$$NX = NX^* = 0$$

G of 4.8 alone to achieve $Y_f = 125$

Budget $T - G = 10 - 13 = -3$ This budget deficit is smaller than the budget deficit of 4.8

(e) Fiscal coordination is difficult to achieve b/c of the benefits of doing nothing and waiting for another economy to undertake a fiscal expansion, as indicated in part (c).